Aquifers, reservoirs, wetlands

SWAT+ Documentation

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Purpose

`aquifer.aqu` contains the properties of the one-dimensional aquifers: initial water table, specific yield, baseflow recession, deep seepage, revap and groundwater nitrate and phosphorus. Each aquifer object in `aquifer.con` selects a row by number in its `aqu` column. The aquifer line of `file.cio` names two files, `initial.aqu` and `aquifer.aqu` (in that order); either can be `null`.

A model has one aquifer object per row of `aquifer.con`. Usually there is a shallow aquifer for each land area and one or more deep aquifers that receive the seepage of the shallow ones. Models that simulate groundwater with the gwflow grid model use `gwflow.con` instead of these aquifers.

`initial.aqu` lists named sets of initial-condition files. In 62.0.1 its rows are read but not used; initial pesticides, pathogens, salts and constituents of aquifers come from `initial.aqu_cs`, described below.

aquifer.aqu

Free format, whitespace-separated. Line 1 is a title and line 2 the column header; both are skipped. Each later line is one aquifer property set with 18 values in this order. The reader does not match columns by header name.

#	Column	Units	Built-in value	Typical range	Description
1	id	-	-	-	Number of the property set. The aqu column of aquifer.con refers to the row by this number.
2	name	-	-	-	Name of the aquifer. Not used by the model.
3	init	-	-	-	Name of an initial-condition set, matched against initial.aqu_cs. See initial.aqu_cs.
4	gw_flo	mm	0.05	0 to 2	Return flow on the day before the simulation starts. It starts the recession, and sets the water volume for the initial nitrate mass.
5	dep_bot	m	0	0 to 10	Depth from the mid-slope surface to the bottom of the aquifer. Must be larger than dep_wt.
6	dep_wt	m	0	0 to 10	Initial depth from the mid-slope surface to the water table.
7	no3_n	mg N/L	0	0 to 1000	Initial nitrate-N concentration.
8	sol_p	mg P/L	0	0 to 1000	Mineral P concentration of the water entering the stream, constant through the run.
9	carbon	%	0.5	0 to 15	Organic carbon content. Reported in the aquifer output; not used in any calculation in 62.0.1.
10	flo_dist	m	50	0 to 1000	Average flow distance to the stream. Not used.
11	bf_max	mm	0	0 to 2	Return flow at which all channels linked to the aquifer receive groundwater. Used only when aquifer flow is distributed to channels with aqu_cha.lin.
12	alpha_bf	1/day	0	0 to 1	Baseflow recession constant.
13	revap	-	0	0 to 1	Revap coefficient: revap as a fraction of potential evapotranspiration.
14	rchg_dp	fraction	0	0 to 1	Fraction of the day's recharge that leaves the aquifer as deep seepage.
15	spec_yld	m^3/m^3	0	0 to 0.4	Specific yield: the volume of water drained by gravity per volume of aquifer. Must be above 0.
16	hl_no3n	days	30	0 to 200	Half-life of nitrate in the aquifer.
17	flo_min	m	0	0 to 10	Water table depth for return flow: the aquifer sends return flow only while the water table is at this depth below the surface or shallower.

#	Column	Units	Built-in value	Typical range	Description
18	`revap_min`	m	0	0 to 10	Water table depth for revap: revap occurs only while the water table is shallower than this depth.

The built-in values apply when aquifer.con points to a number between 1 and the largest id that has no row in the file. They are not filled in for missing columns.

Give every row an id, and keep the ids unique. The ids do not have to be consecutive.

Aquifer water balance

Each day, for each aquifer, in this order. Depths of water are in mm over the aquifer area (the area column of aquifer.con).

Storage and water table

At the start of the run the stored water is

$$S_0 = 1000\,(\text{dep_bot} - \text{dep_wt})\ \text{spec_yld}$$

Each day the recharge R, all water routed into the aquifer from HRUs, routing units and other aquifers, is added to storage, and withdrawals for water allocation are then taken. Irrigation drawn from the aquifer by decision tables is removed from storage when it is applied. The water table depth follows from the storage:

$$h = \max\left(0,\ \text{dep_bot} - \frac{S}{1000\ \text{spec_yld}}\right)$$

The aquifer calculation in 62.0.1 has no recharge delay: recharge is added on the day it arrives.

Return flow

When $h \le$ flo_min, return flow follows the baseflow recession

$$Q_d = Q_{d-1}\,e^{-\alpha} + R_d\,(1 - e^{-\alpha})$$

limited to between zero and the storage, where α is alpha_bf. When $h >$ flo_min, $Q_d = 0$, and the next day's recession starts from zero.

The recession constant is a direct index of how fast groundwater flow responds to recharge (Smedema and Rycroft, 1983). Values are about 0.1 to 0.3 for land with a slow response and 0.9 to 1.0 for land with a rapid response. It is best estimated from measured streamflow during periods without recharge. When the number of baseflow days BFD (the days for baseflow to decline through one log cycle) is known,

$$\alpha = \frac{1}{BFD}\ln 10 = \frac{2.3}{BFD}$$

Return flow leaves through the outflows of type tot in aquifer.con.

Deep seepage

$$\text{seep}_d = \text{rchg_dp}\ R_d$$

limited to the storage. Seepage leaves through the outflows of type rhg in aquifer.con, usually to a deep aquifer. Without such an outflow it leaves the model.

Revap

When $h <$ `revap_min`,

$$\text{revap}_d = \text{revap}\ PET_d$$

limited to the storage, where PET_d is the potential evapotranspiration at the aquifer's weather station (mm). Revap represents water that moves from the aquifer into the unsaturated zone above it when that zone is dry, and uptake by deep-rooted plants. It is removed from the aquifer and not added to the soil. As `revap` approaches 0, the transfer is restricted; as it approaches 1, revap approaches the potential evapotranspiration.

Nitrate and phosphorus

Nitrate recharge is added to the stored nitrate N (kg/ha). The concentration is $c = N/S$, and the nitrate leaving with return flow and seepage is $c\,Q$ and $c\,\text{seep}$. Each day the remaining nitrate decays by

$$N \leftarrow N\ \exp\left(\frac{-0.693}{\text{hl_no3n} + 0.1}\right)$$

The decay represents uptake by bacteria and chemical conversion where the aquifer is depleted of oxygen; it is the net loss from all processes.

The initial nitrate mass is $10\,\text{gw_flo}\,A\,\text{no3_n}/1000$, the concentration times the volume of one day's initial return flow (A in ha). This is a mass in kg for the whole aquifer, but the daily calculation treats the store as kg/ha, so the initial store scales with the aquifer area. It does not use the stored water S_0.

Soluble P in the return flow is sol_p times the day's recharge volume (m^3), divided by 1000, in kg. The load therefore follows the recharge. It appears on days with recharge even when there is no return flow, and is zero on days without recharge. The aquifer stores no phosphorus between days.

Pesticides, salts and constituents

Pesticides entering with recharge are added to the aquifer store and decay with the aquifer half-life of `pesticide.pes` (adjusted to temperature when the temperature decay option is on). The concentration is the stored mass divided by $f\,S + Q + \text{revap} + \text{seep}$, where f is the pesticide groundwater mixing factor of `parameters.bsn`. Return flow and seepage take pesticide out at that concentration, all of it dissolved. Revap takes none.

Salts and other constituents (see `salt` and `cs`) are mixed in the stored water and leave with return flow and seepage at the stored concentration.

Aquifer flow to channels with aqu_cha.lin

When `aqu_cha.lin` links aquifers to channels (see `file.cio`), the return flow is spread over the linked channels as each channel is routed: on the same day when the aquifer comes before the channel in the routing order, otherwise on the next day. The length of channel that receives flow is the total linked channel length times Q/bf_max; when $Q \geq$ `bf_max` all linked channels receive flow, and when it is lower only part of the channel length does, each channel in proportion to its length. Set `bf_max` to a value above zero whenever `aqu_cha.lin` is used.

Calibration and decision tables

These parameters can be changed during calibration through `calibration.cal` (see parameter change files): `flo_init_mm` (`gw_flo`), `dep_bot`, `dep_wt_init` (`dep_wt`), `no3_init` (`no3_n`), `minp_init` (`sol_p`), `cbn_init` (`carbon`), `flo_dist`, `bf_max`, `alpha` (`alpha_bf`), `revap_co` (`revap`), `deep_seep` (`rchg_dp`), `sp_yld` (`spec_yld`), `hlife_n` (`hl_no3n`), `flo_min` and `revap_min`.

Calibration changes are applied after the aquifers are initialised. A change to `sp_yld` recalculates the storage from the current water table, and a change to `no3_init` recalculates the nitrate mass. Changes to `flo_init_mm` and `dep_wt_init` come too late to change the initial state and have no effect. A change to `dep_bot` does not change the stored water; it moves the water table that is calculated from it.

The decision table condition `aqu_dep` tests the water table depth of an aquifer (m). See decision tables.

initial.aqu

Named sets of initial-condition files, in the same layout as `initial.cha`. Two header lines are skipped, then one set per line. The Editor adds a trailing `description` value, which the reader ignores.

#	Column	Description
1	`name`	Name of the set.
2	`org_min`	Name of an organic-mineral initial record.
3	`pest`	Name of a pesticide initial record.
4	`path`	Name of a pathogen initial record.
5	`hmet`	Name of a heavy metal initial record.
6	`salt`	Name of a salt initial record.

In 62.0.1 the rows are read and nothing is done with them: the `init` column of `aquifer.aqu` is not matched against this file, and the aquifer's initial water and nitrate come from `aquifer.aqu`.

initial.aqu_cs

An optional file with a fixed name, read when present. Two header lines are skipped, then one row per line with six names: `name`, `pest`, `path`, `hmet`, `salt` and `cs`.

- `pest` names a row in `pest_water.ini`. The initial pesticide in the aquifer (kg/ha) is the water concentration (mg/kg) times an assumed bulk density of 2.0 t/m³ times `dep_bot` times 10.
- `path` names a row in `path_water.ini`.
- `hmet` is read but not used.
- `salt` names a row in `salt_aqu.ini`: initial salt ion concentrations (g/m³) and salt mineral fractions.
- `cs` names a row in `cs_aqu.ini`: initial constituent concentrations (g/m³) and sorbed concentrations (mg/kg).

Pesticide and pathogen rows are selected by matching `name` with the `init` column of `aquifer.aqu`. Salt and constituent rows are selected by position instead: aquifer 3 (the third row of `aquifer.con`) uses row 3 of `initial.aqu_cs`, whatever its name. Salt and constituent masses are the concentrations times the initial stored water.

Example

`refdata/Osu_1hru` in the SWAT+ repository:

```
aquifer.aqu: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
      id  name                          init        gw_flo       dep_bot        dep_wt         no3_n
       1  aqu011                    initaqu1       0.05000      10.00000       6.00000       0.00000
       2  aqu012                    initaqu1       0.05000      10.00000       6.00000       0.00000
      21  aqu_deep010               initaqu1       0.00000      10.00000      20.00000       0.00000
```

(three of the 21 rows)

The shallow aquifers start with the water table 6 m below a 10 m deep aquifer, so they contain $1000 \times (10-6) \times 0.05 = 200$ mm of water. Return flow and revap both need the water table within 5 m of the surface, so neither occurs until recharge raises it by 1 m (50 mm of storage). The recession constant of 0.95 1/day gives a fast response. One percent of recharge goes to deep seepage. `hl_no3n` is 0, which gives a daily nitrate decay factor of $e^{-6.93}$, so nearly all nitrate entering the aquifer is lost on the day it arrives.

The deep aquifer `aqu_deep010` has `dep_wt` = 20 m, below its `dep_bot` of 10 m, so its initial storage is negative ($1000 \times (10 - 20) \times 0.03 = -300$ mm) and the water table starts 20 m down. It is recharged by the seepage the shallow aquifers send to it. With `flo_min` = 0 it sends return flow only when the water table reaches the surface, and with `revap_min` = 0 it has no revap.

```
initial.aqu: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                      org_min             pest             path             hmet             salt
initaqu1                  no_init             null             null             null             null
```

Related files

- aquifer.con: the aquifer objects, their areas, the `aqu` pointer to `aquifer.aqu` and the `tot` and `rhg` outflows.
- file.cio: the aquifer line, and aqu_cha.lin on the link line.
- pest_water.ini, path_water.ini, salt_aqu.ini, cs_aqu.ini: initial constituents referenced from `initial.aqu_cs`.
- gwflow: the grid groundwater model used instead of these aquifers.
- Outputs: aquifer output files (`aquifer_day` and others).
- Aquifers in the model theory.

Sources

- File name slots: `type input_aqu` in input_file_module.f90, read in readcio_read.f90; read order in proc_aqu.f90
- Readers: aqu_read.f90 (`aquifer.aqu`), aqu_read_init.f90 (`initial.aqu`), aqu_read_init_cs.f90 (`initial.aqu_cs`)
- Type definitions: `aquifer_database`, `aquifer_data_parameters`, `aquifer_dynamic`, `aquifer_init_data_char` and `aquifer_init_data_char_cs` in aquifer_module.f90
- Initialisation: aqu_initial.f90
- Daily calculation: aqu_1d_control.f90, called from command.f90
- `aquifer.con` pointer and outflow types: hyd_read_connect.f90, hyd_connect.f90
- Calibration: cal_parm_select.f90; decision tables: conditions.f90, actions.f90

Column	Fortran variable
id	index of aqudb
name	aqudb%aqunm
init	aqudb%aqu_ini
gw_flo	aqudb%flo
dep_bot	aqudb%dep_bot
dep_wt	aqudb%dep_wt
no3_n	aqudb%no3
sol_p	aqudb%minp
carbon	aqudb%cbn
flo_dist	aqudb%flo_dist
bf_max	aqudb%bf_max
alpha_bf	aqudb%alpha
revap	aqudb%revap_co
rchg_dp	aqudb%seep
spec_yld	aqudb%spyld
hl_no3n	aqudb%hlife_n
flo_min	aqudb%flo_min
revap_min	aqudb%revap_min

`aqudb` is copied to `aqu_dat(iaq)` for each aquifer object in `aqu_initial`; `initial.aqu` rows go to `aqu_init_dat_c`, `initial.aqu_cs` rows to `aqu_init_dat_c_cs`.

Purpose

Reservoirs are impoundments on the channel network. They receive water, sediment and nutrients from upstream objects and release to downstream objects through a release rule. Ponds that are routed as separate objects use the same files. The reservoir group has six files, named on the `reservoir` line of `file.cio` in this order:

Slot	Default name	Content
1	`initial.res`	Named initial-condition sets for reservoirs and wetlands
2	`reservoir.res`	One row per reservoir, naming the records below
3	`hydrology.res`	Reservoir hydrology: spillway areas and volumes, seepage, evaporation, outflow smoothing
4	`sediment.res`	Sediment settling parameters, shared with wetlands
5	`nutrients.res`	Nutrient settling parameters, shared with wetlands
6	`weir.res`	Weir discharge parameters, used by `weir` release rules and by wetlands

Slots 7 and 8 of the same line name the wetland files `wetland.wet` and `hydrology.wet`. A slot set to `null`, or a file that does not exist, leaves that table empty.

`reservoir.res` refers to the other files by name. The release rule named in `reservoir.res` is a decision table in `res_rel.dtl` (see decision tables) or, for names starting with `ctbl_`, a conditions table in `res_conds.dat`.

The five databases are read in every run. `reservoir.res` is read, and the reservoirs are set up, only when `object.cnt` gives one or more reservoirs. `weir.res` is read before the decision tables, so that `weir` actions in `res_rel.dtl` can find their weir record.

All files are free format and whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. The readers do not match columns by header name, so every row needs all of its values in the order given below. Columns after the last one listed (for example the `description` column the SWAT+ Editor writes) are ignored. Names must match, including case.

reservoir.res

One row per reservoir.

#	Column	Units	Description
1	`id`	-	Reservoir number. The row with `id` n is used for reservoir n, the nth reservoir in `reservoir.con`. Numbers must run from 1 to the number of rows.
2	`name`	-	Reservoir name. Not used by the model.
3	`init`	-	Name of a row in `initial.res`.
4	`hyd`	-	Name of a row in `hydrology.res`.
5	`rel`	-	Release rule. A decision table name in `res_rel.dtl`, or a name starting with `ctbl_` for a conditions table in `res_conds.dat`.
6	`sed`	-	Name of a row in `sediment.res`.
7	`nut`	-	Name of a row in `nutrients.res`.

A name that is not found is reported in `diagnostics.out` as `<name> not found (res-hyd)`, `(res-release)`, `(res-sed)` or `(res-nut)`, and the reservoir keeps zero or built-in values for that part. An unmatched `rel` gives an empty release table, so the reservoir releases no water through its rule. An unmatched `init` is not reported; the reservoir then starts from the first row of `om_water.ini`.

hyd, **sed** and **nut** are taken from the row whose **id** equals the reservoir number. **init** and **rel** are taken from the row named by the reservoir's parameter pointer in column 8 of reservoir.con. The SWAT+ Editor writes the same number in both places, so the two agree in Editor-built projects.

initial.res

Named initial-condition sets, shared by reservoirs (**init** in **reservoir.res**) and wetlands (**init** in wetland.wet).

#	Column	Units	Description
1	name	-	Name of the initial-condition set.
2	org_min	-	Name of a row in om_water.ini: initial water volume and concentrations.
3	pest	-	Name of a row in pest_water.ini: initial pesticides in water and benthic sediment.
4	path	-	Name of a pathogen initialization row.
5	hmet	-	Heavy-metal initialization. Read; not used.
6	salt	-	Salt initialization. Read; not used. Reservoir and wetland salts come from the salt input files.

org_min, **pest** and **path** names that are not found fall back to the first row of the referenced file, without a message.

The initial water volume is the **flo** value of the om_water.ini row, a fraction of the principal spillway volume. The concentrations in that row are converted to masses with this volume. A reservoir that is not yet operational at the start of the run (see Operation start) starts empty instead.

hydrology.res

One row per hydrology record.

#	Column	Units	Default	Description
1	name	-	-	Name of the hydrology record, matched against **hyd** in **reservoir.res**.
2	yr_op	year	-	Calendar year in which the reservoir becomes operational. Whole number.
3	mon_op	month	-	Month of **yr_op** in which the reservoir becomes operational. Whole number.
4	area_ps	ha	0.08 × **vol_ps**	Surface area at the principal spillway.
5	vol_ps	ha-m	0.9 × **vol_es**, or 60000	Volume of water needed to fill the reservoir to the principal spillway.
6	area_es	ha	1.5 × **area_ps**	Surface area at the emergency spillway.
7	vol_es	ha-m	1.11 × **vol_ps**	Volume of water needed to fill the reservoir to the emergency spillway.
8	k	mm/hr	-	Hydraulic conductivity of the reservoir bottom. Use 0 for no seepage. Typical range 0 to 1.
9	evap_co	-	0.6	Evaporation coefficient, a multiplier on potential evapotranspiration. Typical range 0 to 1.
10	shp_co1	-	-	Smoothing factor for rising outflow. See Outflow smoothing.
11	shp_co2	-	-	Smoothing factor for falling outflow. See Outflow smoothing.

The Default column gives the value the reader puts in when the column is zero or negative. The rules are applied in this order: if **vol_ps** is not positive, it becomes 0.9 × **vol_es** when **vol_es** is positive and 60000 ha-m otherwise; then **vol_es**, **area_ps**, **area_es** and **evap_co** are filled in as listed. The **area_ps** default corresponds to a mean depth of 12.5 m at the principal spillway.

Principal spillway values describe the normal storage of the reservoir, and emergency spillway values its maximum storage. The volumes are converted from ha-m to m³ (× 10000) when the reservoirs are set up.

A reservoir whose `hyd` name is not found has zero spillway areas and volumes.

Operation start

A reservoir is operational from month `mon_op` of calendar year `yr_op`. Before then it passes its inflow straight through to its downstream objects and keeps no storage. `yr_op` is compared with the calendar year, so a value of 0, or any year before the start of the simulation, makes the reservoir operational from the first day. Only reservoirs that are operational on the first day get the initial storage from `initial.res`; one that starts later begins empty.

Surface area

Each day the surface area A (ha) is calculated from the storage V (m³):

$$A = \beta_1 \, V^{\beta_2}$$

The coefficients are set once at the start of the run from the spillway values, with volumes in m³. When `area_es` is larger than `area_ps` and `vol_es` is larger than `vol_ps`,

$$\beta_2 = \frac{\log_{10} \text{area_es} - \log_{10} \text{area_ps}}{L}, \qquad \beta_1 = \left(\frac{\text{area_es}}{\text{vol_es}} \right)^{\beta_2}$$

where L is the whole-number part of $\log_{10} \text{vol_es} - \log_{10} \text{vol_ps}$. When L is 0 the divisor 0.001 is used instead. β_2 is limited to 0.9, and when the limit applies $\beta_1 = (\text{area_ps}/\text{vol_ps})^{0.9}$. In all other cases $\beta_2 = 0.9$ and $\beta_1 = (\text{area_ps}/\text{vol_ps})^{0.9}$.

In 62.0.1 L is 0 unless the emergency volume is at least ten times the principal volume. For almost every reservoir, then, $\beta_2 = 0.9$ and the relation reduces to

$$A = \left(\text{area_ps} \, \frac{V}{\text{vol_ps}} \right)^{0.9}$$

At the principal spillway this curve gives area_ps$^{0.9}$: 63 ha for an `area_ps` of 100 ha. The surface area sets precipitation on the reservoir, evaporation and seepage, so `area_ps` and `area_es` act on the water balance only through this relation.

Daily water balance

On each day of operation the reservoir:

1. adds the inflow from upstream objects to storage;
2. calculates the release from its release rule, using the storage after inflow, and smooths it (decision tables only);
3. settles sediment (decision tables only);
4. adds precipitation $P = 10 \, R \, A$ (m³, with R the day's precipitation in mm);
5. subtracts the release, limited to the water in storage;
6. subtracts evaporation $E = 10$ evap_co PET A (m³, with PET the day's potential evapotranspiration in mm), limited to storage;
7. subtracts seepage $S = 240 \, k \, A$ (m³, k in mm/hr), limited to storage;
8. updates the surface area from the new storage and calculates nutrient settling and the nutrient load in the release.

Precipitation and potential evapotranspiration come from the weather station assigned to the reservoir in `reservoir.con`. When gwflow is active, seepage is exchanged with the groundwater model and `k` is not used (see gwflow).

Outflow smoothing

For release by decision table, the release Q is smoothed against the previous day's release Q_{d-1} to soften jumps when the rule switches between alternatives:

$$Q_d = \alpha\, Q + (1-\alpha)\, Q_{d-1}, \qquad \alpha = \begin{cases} e^{-\text{shp_co1}} & Q > Q_{d-1} \\ e^{-\text{shp_co2}} & \text{otherwise} \end{cases}$$

With both factors at 0 there is no smoothing. Larger values give more weight to the previous day. The columns keep the names `shp_co1` and `shp_co2` from earlier versions, where they were surface-area shape coefficients. In 62.0.1 the shape coefficients are always calculated by the model as described above and cannot be entered.

Release rules

The `rel` column of `reservoir.res` selects how the reservoir releases water.

Decision tables

A decision table in `res_rel.dtl` evaluates its conditions each day (storage relative to the principal or emergency volume, month, inflow and others; see decision tables) and applies the `release` actions of the alternatives that are met. The `option` field of each action gives the method. Releases from several actions that apply on the same day are added.

`option`	Release per day
`rate`	`const` m³/s.
`rate_pct`	`const` percent of the principal volume.
`inflo_rate`	The larger of inflow plus `const2` m³/s and `const` m³/s.
`inflo_frac`	Inflow multiplied by `const`.
`ab_emer`	All storage above the emergency volume.
`days`	(storage minus a base volume) / `const`, where `const` is the number of drawdown days. The base volume is 0 when `fp` is `null`, `const2` × principal volume when `fp` is `pvol`, and `const2` × emergency volume when `fp` is `evol`.
`dyrt`	(storage minus base volume) / `const` plus `const2` percent of the principal volume. The base volume is 0, the principal volume or the emergency volume for `fp` = `null`, `pvol` or `evol`.
`dyrt1`	As `dyrt`, with the base volume taken from condition 5, 4 or 3 of the table for `fp` = `con1`, `con2` or `con3`.
`inflo_targ`	Inflow plus storage above a target, where the target is the principal volume multiplied by the limit constant of the condition numbered `const`.
`weir`	Weir equation with the `weir.res` row named in `fp`. See weir.res.
`meas`	Measured outflow from the recall record named in `fp` (daily, monthly or annual).
`natlake`	Natural lake outflow after Döll et al. (2003): $Q = k_r\, S_e\, \left(S_e/(V_p - S_0)\right)^{1.5}$ with dead storage S_0 = `const` × principal volume, S_e the storage above S_0, V_p the principal volume and k_r = `const2` (1/day).
`nonirr-h06`, `irr-h06`	Hanazaki et al. (2006) operating rules for non-irrigation and irrigation reservoirs.
`hydrop`	Hydropower rule after Arheimer et al. (2019).

For `days`, the base volume is `const2` times the spillway volume, so an action with `fp` set to `pvol` and `const2` set to 0 draws the reservoir down toward empty. Set `const2` to 1 to draw down to the principal volume.

The `nonirr-h06`, `irr-h06` and `hydrop` options base the release on a five-year record of monthly mean inflow (and, for `irr-h06`, irrigation demand) that each reservoir keeps during the run. In 62.0.1 this record is never filled and stays at zero, so `nonirr-h06` and `hydrop` release no water and `irr-h06` gives an undefined release (zero divided by zero).

In sub-daily runs the actions are applied once for each time step of the day. `days`, `dyrt` (its percentage term) and `meas` divide their daily amount by the number of steps; the other options add their full daily amount at every step. The `weir` option replaces the release with one time step's weir flow at each step.

For a reservoir, the `weir` option takes the weir crest height from the wetland data of the HRU with the same number as the reservoir. When that HRU has no wetland the crest height is zero, so the whole water depth (storage divided by surface area) acts as head. The `height` column of `weir.res` and the `weir_height` decision table action do not change the crest height that the reservoir weir release uses.

Conditions tables

A `rel` name starting with `ctbl_` is looked up in `res_conds.dat`, a file with a fixed name that is read when it exists in the run directory. The release is the value given by the matching condition of the table. Outflow smoothing and sediment settling are not applied to these reservoirs, and their release contains no sediment.

sediment.res

One row per sediment record. Reservoirs refer to it through `sed` in `reservoir.res`, wetlands through `sed` in `wetland.wet`.

#	Column	Units	Description
1	`name`	-	Name of the sediment record.
2	`sed_amt`	mg/L	Equilibrium (normal) suspended sediment concentration. Settling occurs only above this concentration. Typical range 1 to 5000.
3	`d50`	µm	Median particle size of suspended and benthic sediment. Sets the settling rate.
4	`carbon`	%	Organic carbon in suspended and benthic sediment. Used in pesticide partitioning.
5	`bd`	t/m³	Bulk density of benthic sediment. Used in pesticide partitioning between water and benthic sediment.
6	`sed_stl`	-	Sediment settling rate. Read but not used.
7	`stl_vel`	m/day	Sediment settling velocity. Read but not used.

Sediment settling

Each day, when the sediment concentration C (mg/L) in the reservoir is above the equilibrium concentration C_{eq} (`sed_amt`), the excess settles at a rate set by the median particle size:

$$C' = (C - C_{eq})\, e^{-0.184\, d_{50}} + C_{eq}$$

With `d50` at 10 µm, 16% of the excess concentration remains after one day; at 2 µm, 69% remains. The release takes sediment at the concentration C', and all silt, clay, sand and aggregate fractions in the reservoir settle.

When the concentration is at or below `sed_amt`, nothing settles and the release takes no sediment that day.

Median particle sizes of sediment classes: clay 1.95 µm, fine silt 4.9 µm, medium silt 11.7 µm, coarse silt 47 µm, fine sand 94 µm, medium sand 375 µm, coarse sand 1500 µm. A reservoir receives sediment from its whole upstream area, so a representative value for that area suits it better than the texture of one HRU.

nutrients.res

One row per nutrient record. Reservoirs refer to it through `nut` in `reservoir.res`, wetlands through `nut` in `wetland.wet`.

#	Column	Units	Description
1	`name`	-	Name of the nutrient record.
2	`mid_start`	month	First month of the mid-year settling period. Whole number, 1 to 12.
3	`mid_end`	month	Last month of the mid-year settling period. Whole number, 1 to 12.

#	Column	Units	Description
4	`mid_n_stl`	m/year	Organic nitrogen settling rate in the mid-year period.
5	`n_stl`	m/year	Organic nitrogen settling rate in the rest of the year.
6	`mid_p_stl`	m/year	Particulate phosphorus settling rate in the mid-year period.
7	`p_stl`	m/year	Particulate phosphorus settling rate in the rest of the year.
8	`chla_co`	m/year	Loss rate for soluble nitrogen (nitrate, ammonium and nitrite).
9	`secchi_co`	m/year	Loss rate for soluble phosphorus.
10	`theta_n`	-	Temperature adjustment factor for nitrogen losses.
11	`theta_p`	-	Temperature adjustment factor for phosphorus losses.
12	`n_min_stl`	mg/L	Nitrogen concentration below which no nitrogen is lost.
13	`p_min_stl`	mg/L	Phosphorus concentration below which no phosphorus is lost.

The Editor headers `chla_co` and `secchi_co` come from older versions, where these columns were a chlorophyll-a coefficient and a water clarity coefficient. In 62.0.1 they are the soluble nitrogen and phosphorus loss rates. Chlorophyll-a and Secchi depth are not calculated from them.

The six rates are entered in m/year and divided by 365 when the file is read.

Nutrient losses

The mid-year rates apply in the months from `mid_start` to `mid_end`, and the other rates in the remaining months. The test is `mid_start` ≤ month ≤ `mid_end`, so a period that runs across the end of the year (for example November to March) never applies and the other rates are used all year. To use one rate all year, give both columns the same value.

Each day the fraction of organic nitrogen lost is

$$k_N = \frac{\text{mid_n_stl}}{365}\,\theta_N^{T-20}\,(C_N - C_{N,min})$$

limited to between 0 and 1, where C_N is the organic nitrogen concentration (mg/L), $C_{N,min}$ is `n_min_stl`, θ_N is `theta_n` and T is the day's mean air temperature (°C) at the weather station. Particulate phosphorus is treated the same way with `mid_p_stl` or `p_stl`, `theta_p` and `p_min_stl`. The organic nitrogen and particulate phosphorus masses are multiplied by $1 - k$.

Soluble nitrogen uses `chla_co` and the total soluble nitrogen concentration, and soluble phosphorus uses `secchi_co`. For these, the loss fraction is multiplied by a fixed 0.2 before it is applied to nitrate, ammonium, nitrite and soluble phosphorus.

The rate is multiplied by a concentration difference and the result is used as a fraction, so the units of the product do not match: a value of 1 m/year acts as a daily fraction of 1/365 per mg/L of excess concentration. The fraction cannot go below zero, so negative rates give no loss and the sediment never becomes a source of nutrients.

The release takes each nutrient in proportion to its share of the water: release / (storage + release).

Measured apparent settling velocities for phosphorus in natural lakes most often lie between 5 and 20 m/year, with reported values from below 1 to over 200 m/year (Chapra, 1997). Reservoirs show a wider range: Higgins and Kim (1981) reported −90 to 269 m/year for 18 reservoirs in Tennessee (median 42.2 m/year), and Walker and Kiihner (1978) reported −1 to 125 m/year for 27 Midwestern reservoirs (mean 12.7 m/year). Panuska and Robertson (1999) recommend these values:

Water body nutrient dynamics	Settling velocity (m/year)
Shallow, high net internal phosphorus flux	0 or less
Moderate net internal phosphorus flux	1 to 5
Minimal net internal phosphorus flux	5 to 16

Water body nutrient dynamics	Settling velocity (m/year)
High net internal phosphorus removal	more than 16

In 62.0.1 values of 0 or less give no phosphorus loss.

weir.res

One row per weir record. A reservoir uses a weir record through a `weir` action in its release decision table (the record named in the action's `fp` field). Wetlands use weir records as described under storage volumes and weir height on the wetland page.

#	Column	Units	Built-in value	Description
1	`name`	-	-	Name of the weir record.
2	`linear_c`	-	1.84	Weir discharge coefficient c.
3	`exp_k`	-	2.6	Weir discharge exponent k.
4	`width`	m	2.5	Weir width w.
5	`height`	m	0	Height of the weir crest above the bottom of the impoundment. Used for wetlands only.

The built-in values apply when a `weir` action names a record that is not in the file.

The weir discharge is

$$Q = 0.45\, A\, c\, w\, H^{k}$$

in m^3/s, where A is the surface area of the water body (ha) and H is the water depth above the weir crest (m). In daily runs the discharge is applied hour by hour for up to 24 hours, each hour limited to the water above the crest, and the depth is updated after each hour.

Calibration

These reservoir parameters can be changed through `calibration.cal`: `psa`, `pvol`, `esa` and `evol` (spillway areas in ha and volumes in m^3), `k_res` (`k`), `evrsv` (`evap_co`), `nsed` (`sed_amt`), `res_d50` (`d50`), `sed_stlr` (`sed_stl`), `velsetlr` (`stl_vel`), `nsetlr1`, `nsetlr2`, `psetlr1`, `psetlr2` (the four settling rates), `nsolr` (`chla_co`), `psolr` (`secchi_co`), `soln_stl_fr` and `solp_stl_fr` (the fixed 0.2 factors), the storage and constituent masses `vol`, `sed`, `orgn`, `orgp`, `no3`, `nh3`, `no2` and `solp`, and the release-table constants `drawdown_days` (`const`) and `withdraw_rate` (`const2`).

Calibration acts on the values after unit conversion, so settling rates are changed in m/day. Changing `psa`, `pvol`, `esa` or `evol` does not recalculate the surface area coefficients.

Example

`refdata/Osu_1hru` in the SWAT+ repository has no reservoirs. Its sediment, nutrient, weir and initial-condition files are used by its wetland, and show the formats.

`initial.res`:

```
initial.res: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                      org_min             pest             path             hmet             salt
initwet1                  no_init             null             null             null             null
```

`sediment.res`:

```
sediment.res: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                    sed_amt           d50        carbon            bd       sed_stl       stl_vel
sedwet1                   10.00           5.00       0.04            0.8          0.15       0.0002
```

Sediment settles above 10 mg/L. With `d50` at 5 µm, 40% of the excess remains after one day.

`nutrients.res`:

```
nutrients.res: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name               mid_start   mid_end     mid_n_stl         n_stl     mid_p_stl         p_stl       chla_c
nutwet1                    5        10       5.50000       5.50000      10.00000      10.00000       1.00000
```

The mid-year period is May to October, but the two rates for each nutrient are equal, so one rate applies all year. Both `theta` values are 1, so the losses do not depend on temperature.

`weir.res`:

```
weir.res: Reservoir weir inputs - asdf;lj
NAME   Linear_C    Exp_K     Width(m)   Depth(m)
weir1   1.83       1.50      5.00       0.0
```

The header line here differs from the Editor's. It is skipped, so only the order of the values matters.

A `hydrology.res` and `reservoir.res` pair for one reservoir, written for illustration:

```
hydrology.res: example
name        yr_op  mon_op   area_ps   vol_ps   area_es   vol_es      k   evap_co  shp_co1  shp_co2
hydres1         0       0     150.0   1500.0     200.0   2200.0   0.05      0.60     0.00     0.00
```

```
reservoir.res: example
id   name      init      hyd       rel             sed      nut
1    res001    initres1  hydres1   drawdown_days   sedres1  nutres1
```

The reservoir is operational from the start. Its principal volume is 1500 ha-m (15 million m³). Because `vol_es` is less than ten times `vol_ps`, the surface area follows $A = (150\,V/1.5 \times 10^7)^{0.9}$, which gives 91 ha at the principal spillway. Outflow is not smoothed. Seepage is $240 \times 0.05 \times A$ m³ per day.

The `drawdown_days` table in `refdata/Osu_1hru/res_rel.dtl`:

```
name                    conds      alts      acts
drawdown_days               2         2         2
var                       obj   obj_num           lim_var            lim_op     lim_const      alt1
vol                       res         0              pvol                 *       1.00000         >
vol                       res         0              evol                 *       1.00000         <
act_typ                   obj   obj_num              name            option         const        const2
release                   res         0         over_prin              days      15.00000       0.00000
release                   res         0    over_emergency              days       5.00000       0.00000
```

Between the principal and emergency volumes the reservoir releases storage / 15 per day; above the emergency volume, storage / 5 per day. Below the principal volume no alternative applies and nothing is released. Because `const2` is 0, the base volume is zero in both actions, and each releases a share of the whole storage.

Related files

- `reservoir.con`: places each reservoir in the routing network and assigns its weather station.
- `object.cnt`: number of reservoirs.
- `file.cio`: names the reservoir files.
- `res_rel.dtl`: release decision tables.
- `om_water.ini`: initial water and concentrations referenced from `initial.res`.
- `recall.rec`: measured outflows for `meas` releases.
- `wetland.wet` and `hydrology.wet`: wetlands, which share `initial.res`, `sediment.res`, `nutrients.res`, `weir.res` and `res_rel.dtl`.
- Reservoir output and reservoir and wetland theory.

Sources

- File names: `type input_res` in input_file_module.f90, read in readcio_read.f90
- Call order: main.f90.in, proc_res.f90
- Readers: res_read.f90, res_read_init.f90, res_read_hyd.f90, res_read_sed.f90, res_read_nut.f90, res_read_weir.f90, res_read_conds.f90, dtbl_res_read.f90
- Type definitions: reservoir_data_module.f90, reservoir_module.f90
- Setup: res_objects.f90, res_allo.f90, res_initial.f90, `res_convert_mass` in hydrograph_module.f90
- Daily routines: res_control.f90, res_hydro.f90, res_rel_conds.f90, res_sediment.f90, res_nutrient.f90, theta.f90, res_pest.f90
- Decision table conditions and actions: conditions.f90, actions.f90
- Calibration: cal_parm_select.f90

File	Column	Fortran variable
reservoir.res	`id`, `name`, `init`, `hyd`, `rel`, `sed`, `nut`	index into `res_dat_c`, then `res_dat_c%name`, `%init`, `%hyd`, `%release`, `%sed`, `%nut`
initial.res	`name`, `org_min`, `pest`, `path`, `hmet`, `salt`	`res_init_dat_c%init`, `%org_min`, `%pest`, `%path`, `%hmet`, `%salt`
hydrology.res	`yr_op`, `mon_op`	`res_hyddb%iyres`, `%mores`
hydrology.res	`area_ps`, `vol_ps`, `area_es`, `vol_es`	`res_hyddb%psa`, `%pvol`, `%esa`, `%evol`
hydrology.res	`k`, `evap_co`	`res_hyddb%k`, `%evrsv`
hydrology.res	`shp_co1`, `shp_co2`	`res_hyddb%br1`, `%br2`, copied to `res_ob%lag_up`, `%lag_down`
sediment.res	`sed_amt`, `d50`, `carbon`, `bd`, `sed_stl`, `stl_vel`	`res_sed%nsed`, `%d50`, `%carbon`, `%bd`, `%sed_stlr`, `%velsetlr`
nutrients.res	`mid_start`, `mid_end`	`res_nut%ires1`, `%ires2`
nutrients.res	`mid_n_stl`, `n_stl`, `mid_p_stl`, `p_stl`	`res_nut%nsetlr1`, `%nsetlr2`, `%psetlr1`, `%psetlr2`
nutrients.res	`chla_co`, `secchi_co`	`res_nut%nsolr`, `%psolr`
nutrients.res	`theta_n`, `theta_p`, `n_min_stl`, `p_min_stl`	`res_nut%theta_n`, `%theta_p`, `%conc_nmin`, `%conc_pmin`
weir.res	`linear_c`, `exp_k`, `width`, `height`	`res_weir%c`, `%k`, `%w`, `%h`

Purpose

A wetland in SWAT+ is an HRU that can store ponded water on its surface. The same mechanism simulates rice paddies. Two files are specific to wetlands, named in slots 7 and 8 of the `reservoir` line of file.cio:

- `wetland.wet`: one row per wetland, naming its initial conditions, hydrology, release rule, sediment and nutrient records.
- `hydrology.wet`: wetland hydrology records. Depths and areas are relative to the HRU.

Wetlands also use four reservoir files, described on the reservoir page: initial.res, sediment.res, nutrients.res and weir.res. Release rules are decision tables in res_rel.dtl (see decision tables).

An HRU becomes a wetland when the `surf_stor` column of hru-data.hru names a row of `wetland.wet`. With `null` there, the HRU has no ponded storage.

Both files are free format and whitespace-separated. Line 1 is a title and line 2 is the column header; both are skipped. The readers do not match columns by header name, so every row needs all of its values in the order given below. Columns after the last one listed are ignored. Names must match, including case. A slot in file.cio set to `null`, or a file that does not exist, leaves the table empty.

wetland.wet

One row per wetland.

#	Column	Units	Description
1	`id`	-	Wetland number. Read; not used. Rows are used in the order they appear.
2	`name`	-	Wetland name, matched against `surf_stor` in `hru-data.hru`.
3	`init`	-	Name of a row in `initial.res`.
4	`hyd`	-	Name of a row in `hydrology.wet`. The name `paddy` switches on rice paddy behaviour; see Outflow.
5	`rel`	-	Name of a release decision table in `res_rel.dtl`. Not used for paddies.
6	`sed`	-	Name of a row in `sediment.res`.
7	`nut`	-	Name of a row in `nutrients.res`.

Several HRUs can use the same row. A name that is not found is reported in `diagnostics.out` as `<name> not found (wet-init)`, `(wet-hyd)`, `(wet-release)`, `(wet-sed)` or `(wet-nut)`, and a `surf_stor` name that is not in `wetland.wet` as `<name>not found (wetland.wet)`. A wetland whose `hyd` is not found uses the built-in hydrology values (zero depths and areas). A non-paddy wetland whose `rel` is not found releases no water through its rule.

Some files, such as the example below, have an eighth column naming a weir record. 62.0.1 does not read it; how a wetland gets its weir record is described under Storage volumes and weir height.

hydrology.wet

One row per wetland hydrology record.

#	Column	Units	Built-in value	Description
1	`name`	-	-	Name of the hydrology record, matched against `hyd` in `wetland.wet`.
2	`hru_ps`	fraction	0	Fraction of the HRU area covered at the principal spillway (when flow over the surface inlet riser starts). Used only in the surface area set up before the first day, which the daily calculation replaces.
3	`dp_ps`	mm	0	Average water depth at the principal spillway. Sets the principal volume and, without weir records, the weir height.
4	`hru_es`	fraction	0	Fraction of the HRU area covered at the emergency spillway (when water starts to spill into the ditch). Not used in the daily simulation.
5	`dp_es`	mm	0	Average water depth at the emergency spillway. Sets the emergency volume.
6	`k`	mm/hr	0.01	Hydraulic conductivity of the wetland bottom. Sets the seepage rate.
7	`evap`	-	0.7	Evaporation coefficient, a multiplier on potential evapotranspiration for ponded water.
8	`vol_area_co`	-	1	Coefficient a of the volume–area relation.
9	`vol_dp_a`	-	1	Coefficient b of the volume–depth relation.
10	`vol_dp_b`	-	1	Coefficient c of the volume–depth relation.

#	Column	Units	Built-in value	Description
11	`hru_frac`	fraction	0.5	Fraction of the HRU that drains into the wetland. Read but not used.

The built-in values apply to wetlands whose `hyd` name is not found. In rows of the file, the reader replaces zero or negative values in four columns:

- `evap` becomes 0.6.
- `k` becomes the saturated hydraulic conductivity of the top soil layer of the HRU whose number equals the row number of the record in `hydrology.wet` (HRU 1 for the first row, and so on), which is not in general an HRU that uses the record.
- `hru_ps` and `hru_es` are set to 0.08 × and 1.5 × values that are still zero when the file is read, so they stay at 0.

Wetland behaviour

Storage volumes and weir height

At the start of the run, and again when a decision table switches the wetland on, the storage limits are set from the HRU area A_{hru} (ha) on the assumption that the whole HRU is ponded:

$$V_p = 10\,A_{hru}\,\mathrm{dp_ps}, \qquad V_e = 10\,A_{hru}\,\mathrm{dp_es}$$

in m³. `hru_ps` and `hru_es` do not enter the volumes.

If `weir.res` has at least one record, every wetland starts with the first record of `weir.res` as its weir: the weir height is that record's `height` (m), the principal volume becomes $V_p = 10^4\,A_{hru}$ height, and V_e is raised to at least $1.2\,V_p$. Without weir records the weir height is `dp_ps` / 1000 m. A `weir` operation in the management schedule can later change the height and select another `weir.res` record by name (see Management and decision tables).

The initial water volume is the `flo` fraction of the `om_water.ini` row named in `initial.res`, multiplied by V_p, with concentrations from the same row.

Daily water balance

For an HRU with a wetland, the daily HRU routines change as follows:

- Surface run-on from upstream objects (with tile flow) is added to the wetland storage.
- The HRU generates no surface runoff. The day's precipitation, over the whole HRU area ($10\,R\,A_{hru}$ m³), and any irrigation water are added to the wetland storage.
- Evaporation from the ponded water is taken before soil evaporation; while water is ponded there is no soil evaporation (see Evaporation).
- Seepage from the ponded water is the infiltration into the soil.
- Water above saturation in the soil profile is returned to the wetland storage. If the weir height is below 1 mm it becomes surface runoff instead.
- The wetland outflow is the surface runoff of the HRU.

Water depth is storage divided by the HRU area: $d = V/(10\,A_{hru})$ mm.

Surface area

For wetlands that are not paddies, the water surface area A_w follows the storage V relative to the principal volume V_p:

$$x = b^2 + 4c\left(1 - \frac{V}{V_p}\right), \qquad h = \frac{-b - \sqrt{x}}{2c} + b, \qquad A_w = A_{hru}\,\min(1, \max(0.01,\ 1 + a\,h))$$

with a = `vol_area_co`, b = `vol_dp_a` and c = `vol_dp_b`. When x is below 10^{-6}, h is 0. With the built-in values ($a = b = c = 1$) the whole HRU is covered at $V = V_p$, and 63% at $V = 0.5\,V_p$. The area set up before the first day also multiplies the fraction by `hru_ps`.

For paddies the surface area is always the HRU area.

The surface area sets seepage and the weir flow. Precipitation, evaporation and the water depth are always calculated over the whole HRU.

Evaporation

When the HRU has ponded water and its leaf area index is 4 or less, the maximum evaporation is

$$E_{max} = \text{evap}\,\left(1 - \frac{LAI}{4}\right) PET$$

With a leaf area index above 4 there is no evaporation from the ponded water. Evaporation is taken from the ponded water first, and from the soil only when no ponded water remains.

In 62.0.1 the `evap` value is looked up through the record's row number in `hydrology.wet` used as an HRU number: the coefficient applied is that of the wetland on the HRU with that number, or 0.7 if that HRU has no wetland. In a model where HRU 1 uses the first `hydrology.wet` record, HRU 2 the second and so on, this gives the intended value.

Seepage

Daily seepage is

$$S = \min\left(V,\ 240\,k\,A_w\right)$$

in m³, with k in mm/hr. When the seepage is more than 0.1 mm over the HRU, it is reduced by the amount the soil profile cannot take before it becomes saturated. Nitrate, ammonium, soluble phosphorus, organic nitrogen and organic phosphorus move with the seepage into the top soil layer in proportion to the water. A puddling operation can replace `k` during the run. When gwflow is active, the exchange with groundwater is calculated by the groundwater model (see gwflow).

Outflow

A wetland whose `hyd` name in `wetland.wet` is `paddy` is a rice paddy. Its outflow is weir flow from the `weir.res` record selected for it, and `rel` is not used:

- If the storage exceeds V_e, the water above V_e is released.
- While the water depth is above the weir height, flow over the weir is $Q = 0.45\,A\,c\,w\,H^k$ (m³/s), with the weir.res coefficients, the HRU area A (ha) and the depth above the weir H (m). It is applied hour by hour for up to 24 hours in a daily run, each hour limited to the water above the weir.
- With the weir height at zero the whole water depth acts as head, so the ponded water drains over the weir.

Other wetlands release through the decision table named in `rel`. The conditions and release options are those of reservoirs (release rules), evaluated against the wetland's storage, principal and emergency volumes, water depth and weir height. The outflow is not smoothed.

Paddy HRUs also write a daily line to `paddy_test.csv` with precipitation, irrigation, seepage, evapotranspiration terms, weir height, water depth, weir outflow, soil water, sediment and nitrate.

Sediment and nutrients

Sediment settling and nutrient losses in the ponded water follow the reservoir equations with the wetland's `sediment.res` and `nutrients.res` records (see sediment settling and nutrient losses), using the HRU's weather. When the wetland was empty at the start of the day and releases water, the sediment yield of the outflow is calculated with a MUSLE-type equation from the HRU's cover and erosion factors, as for a flushed field.

Management and decision tables

These operations and decision-table entries act on wetlands and paddies:

- Decision table conditions `vol_wet` (storage against the principal or emergency volume), `wet_depth` (water depth, mm) and `weirh` (weir height, mm). See decision tables.
- Decision table action `weir_height` with option `wet`: sets the weir height to `const` mm, sets V_p to the HRU area times that height, and raises V_e to at least $1.2\,V_p$. It does not change the weir record.
- Management operation `weir`: sets the weir height to its third data value (mm), sets V_p the same way with V_e raised to at least $1.1\,V_p$, and selects the `weir.res` record named in its first data field.
- Decision table actions `impound_off` and `impound_on`: `impound_off` removes the wetland and its stored water from the HRU. `impound_on` sets the wetland up again from the `surf_stor` row of `hru-data.hru`; the name given in the action's file pointer is not matched in 62.0.1.
- Puddling (`puddle` action, with `puddle.ops`): replaces the wetland bottom conductivity and raises the sediment concentration of the ponded water.
- Fertilizer applied to an HRU with ponded water goes into the water when `carbon` in `codes.bsn` is 2. With other `carbon` values it is counted in the fertilizer output but added to neither the water nor the soil.
- Tillage when the water depth is above 1 mm mixes the soil layers with the ponded-water routine, including pesticides in the soil.

See management schedules and operations for the operation formats.

When gwflow is active and `gwflow.wetland` exists, the wetland bed thickness (m) for each wetland HRU is read from it: two header lines, then one line per wetland with a name and a thickness, where the digits in the name give the HRU number (for example `wet019` for HRU 19).

Example

`refdata/Osu_1hru` in the SWAT+ repository is a single rice paddy HRU. `hru-data.hru` names `paddy0001` in `surf_stor`.

`wetland.wet`:

```
wetland.       wet: written by SWAT+ edi#############for SWAT+ rev.60.5.4
     id                name        init         hyd         rel         sed         nut        WEIR
      1          paddy0001    initwet1       paddy        weir     sedwet1     nutwet1       weir1
```

The title line is damaged but is skipped. The hydrology record is `paddy`, so the HRU is a paddy and the `weir` decision table named in `rel` is not used. The `WEIR` column is not read.

`hydrology.wet`:

```
hydrology.wet: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                    hru_ps         dp_ps        hru_es         dp_es             k          evap    vol
paddy                     1.00        150.00          1.00        150.00           0.5          0.75
```

The emergency volume is 150 mm of water over the HRU. Seepage is at most $0.5 \times 24 = 12$ mm per day. Ponded water evaporates at $0.75 \times$ PET while the rice canopy is sparse, falling to zero as the leaf area index reaches 4.

`weir.res` has one record, `weir1`, with a height of 0 m, so the paddy starts with a weir height of zero and a principal volume of zero, and any ponded water drains over the weir. During the season `weir_height` actions in `lum.dtl` raise the weir to 60 mm and lower it to 0 again. While water stands above 60 mm it flows over the 5 m weir at $Q = 0.45\,A \times 1.83 \times 5 \times H^{1.5}$ m³/s.

Related files

- `hru-data.hru`: assigns a `wetland.wet` row to an HRU through `surf_stor`.
- `file.cio`: names the wetland files.
- `initial.res`, [`sediment.res`](#input-reference-res-sedimentres), [`nutrients.res`](#input-reference-res-nutrientsres) and `weir.res`: shared with reservoirs.

- res_rel.dtl and lum.dtl: wetland release rules and wetland actions.
- management.sch and operations: weir, irrigation, puddling, fertilizer and tillage operations on paddies.
- Salt and constituent inputs for wetlands.
- Water balance output and reservoir and wetland theory.

Sources

- File names: `type input_res` in input_file_module.f90; call order in main.f90.in
- Readers: wet_read.f90, wet_read_hyd.f90
- Type definitions: `type reservoir_data_char_input` and `type wetland_hyd_data` in reservoir_data_module.f90; `type wetland` in reservoir_module.f90
- Setup: hru_allo.f90, wet_all_initial.f90, wet_initial.f90
- Daily routines: hru_control.f90, et_act.f90, wetland_control.f90, res_weir_release.f90, res_hydro.f90, res_sediment.f90, res_nutrient.f90, swr_satexcess.f90
- Conditions, actions and operations: conditions.f90, actions.f90, mgt_sched.f90, pl_fert_wet.f90, mgt_newtillmix_wet.f90

File	Column	Fortran variable
wetland.wet	id	read into a dummy k
wetland.wet	name, init, hyd, rel, sed, nut	wet_dat_c%name, %init, %hyd, %release, %sed, %nut
hydrology.wet	hru_ps, dp_ps, hru_es, dp_es	wet_hyddb%psa, %pdep, %esa, %edep
hydrology.wet	k, evap	wet_hyddb%k, %evrsv
hydrology.wet	vol_area_co, vol_dp_a, vol_dp_b	wet_hyddb%acoef, %bcoef, %ccoef
hydrology.wet	hru_frac	wet_hyddb%frac